Counter-Rotating Aperture Resolution: A Formal Stream-Processing Model for SHA-256 Recurrence, Root-Cover Coupling, and Transformer Language Attention

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Abstract

This paper formalizes **Counter-Rotating Aperture Resolution** (CRAR): a model in which a fixed aperture and a moving data surface produce output through relative phase contact. The central claim is operational rather than metaphoric: a static output, or "still," cannot identify the stream that generated it. A stream is an operator, recurrence, or transition law, and it is exposed only through controlled perturbation across the aperture.

The theory is developed in three stages. First, we define stills, streams, apertures, and counter-rotating domains using a train-and-circle model. A processor may remain externally fixed while an oppositely moving surface passes beneath it; computation occurs at the contact point, not in absolute translation. Second, we

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instantiate the model in the SHA-256 message schedule. The lag set $L = \{2,7,15,16\}$ and seed width $N = 16$ generate the aperture complement

$$A_N(L) = \{N - \ell : \ell \in L\} = \{0,1,9,14\},$$

which organizes the direct-entry taxonomy. The first two expanded words have parent apertures

$$P_{16} = \{0,1,9,14\}, \qquad P_{17} = \{1,2,10,15\},$$

whose unique intersection is $\{1\}$. Thus $W[1]$ is the **Root-Cover Axis**: the unique seed word that directly seeds both expansion roots $W[16]$ and $W[17]$, has $D(1) = \{16,17\}$, has clean echo set $\mathcal{C}_1 = \emptyset$, and saturates immediately at $S_1 = 16$. The $W[9]$ result is then correctly bounded: $W[9]$ is the first triple-entry, length-sensitive face-selection hinge, while $W[1]$ is the global coupling axis.

Third, we translate the mathematics into transformer attention. Causal language models process symbolic streams by passing prompt tokens across fixed weights and a causal mask. Attention is not simple forward reading; it is counter-rotating aperture contact between a forward token stream and a backward-facing context field. We distinguish a **forbidden seam**, implemented by the causal mask $M_{ij} = -\infty$ for $j > i$, from a **balance seam**, such as equal-score or orthogonality surfaces in $QK^{\top}$. The SHA-to-transformer mapping is stated as a testable transport hypothesis, not as a completed proof.

The result is a unified mathematical paper on why the word is the still, meaning is the stream crossing it, and controlled perturbation is the method that reveals the operator.

1. Scope and Thesis

The working principle is:

A still cannot reveal a stream.

A still is a projected state. A stream is the transition law that moves states. If only the final state is observed, the underlying operator is generally non-identifiable. This matters for cryptographic schedules, language models, and any symbolic processor whose visible output hides the internal recurrence.

In SHA-256, the final digest is a still. The message schedule is the stream. In language, a word token is a still. Meaning is the stream crossing the token through context, memory, and prediction. In both cases, the correct object is not the output value, but the preserved transition structure exposed by perturbation.

This paper therefore studies the operator geometry rather than the final output.

This paper does **not** claim a break of SHA-256. It does not claim that transformer attention has been fully proven equivalent to SHA-256 recurrence. It extracts the shared mathematics of aperture, counterflow, recurrence, masking, contamination, saturation, and projection.

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2. Stills, Streams, and Operators

Let S_t denote a system state at discrete time t. A **still** is a projection of a state:

$$y_t = \pi(S_t).$$

A **stream** is the transition law:

$$S_{t+1} = F(S_t),$$

or, more generally,

$$S_{t+1} = F\big(S_t, S_{t-k_1}, \dots, S_{t-k_m}\big).$$

The differential view is

$$\Delta S_t = S_{t+1} - S_t.$$

The still is residue. The stream is recurrence.

2.1 Static Non-Identifiability Lemma

Lemma 2.1. Let $\pi: S \to Y$ be a many-to-one projection. Given only $y_T = \pi(S_T)$, the transition operator F that generated S_T is not uniquely identifiable.

Proof. Since π is many-to-one, there exist distinct states $S_T, S'_T \in S$ such that $\pi(S_T) = \pi(S'_T) = y_T$. There may exist distinct transition operators F and G and initial states S_0, S'_0 satisfying

$$F^T(S_0) = S_T, \qquad G^T(S'_0) = S'_T.$$

Both generate the same projected still y_T. Therefore y_T alone cannot determine the stream operator. ▫

2.2 Perturbation Principle

A stream is exposed by controlled perturbation:

$$\delta S_0 \mapsto \delta S_1 \mapsto \cdots \mapsto \delta S_T.$$

Define a perturbation-response map

$$R_F(t, j) = \mathbf{1}\left[\frac{\partial S_t}{\partial S_j} \neq 0\right]$$

for differentiable systems, or a Boolean contamination recurrence for discrete systems. The operator becomes visible through where motion can and cannot propagate.

Thus:

Truth is not in the still value. Truth is in the preserved transition.

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3. The Train-Circle Model of Counter-Rotating Recursion

Consider a train moving linearly:

$$x_{\text{train}}(t) = vt.$$

On the train is a circular recursive domain of radius r with angular phase

$$\theta(t) = \omega t \ (\text{mod } 2\pi).$$

A point on the circle has global position

$$x(t) = vt + r\cos(\omega t),$$

$$y(t) = r\sin(\omega t).$$

The train moves forward in global time, but the circular domain returns to previous local phase states:

$$\theta(t+T) = \theta(t), \qquad T = \frac{2\pi}{\omega}.$$

Thus the system does not reverse external time; it returns to a previous **domain state**.

3.1 Radius as Phase Curvature

The tangential velocity on the circle is

$$v_\theta = r\omega.$$

At the center, $r = 0$, the point follows the train:

$$x(t) = vt, \qquad y(t) = 0.$$

Near the rim, phase dominates. Therefore:

Distance from the fixed point determines phase curvature.

3.2 Counter-Rotating Surface

Now suppose an observer attempts to walk one way while the circular surface moves the opposite way. Let

$$v_{\text{observer}} + v_{\text{surface}} = 0.$$

Then the observer has no external displacement:

$$\Delta x = 0.$$

But the surface still passes beneath the contact point. The relative velocity is

$$v_{\text{rel}} = v_{\text{observer}} - v_{\text{surface}}.$$

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So processing can occur even while the observer does not move globally. The contact point becomes a read head.

This is the central physical image of CRAR:

Processing is relative motion across a fixed aperture.

4. Counter-Rotating Aperture Resolution

A **counter-rotating aperture system** is a tuple

$$\mathcal{A} = (X, G, M, \alpha, \pi),$$

where:

- $X = (x_0, x_1, \dots)$ is the moving data stream.
- G is the fixed or slowly varying geometric field.
- M is the hard constraint or mask.
- α is the aperture contact operator.
- π is the final projection or readout.

The local contact state is

$$z_t = \alpha(x_t, G, M).$$

The output still is

$$y_t = \pi(z_t).$$

The data stream may move forward while the aperture reaches backward into prior states:

$$z_t = \alpha\left(x_t, x_{t-k_1}, \dots, x_{t-k_m}, G, M\right).$$

This gives the CRAR law:

Forward execution plus backward reference equals recursive time fold.

5. SHA-256 Message Schedule as a Counter-Rotating Aperture

The SHA-256 message schedule expands a 16-word seed window $W[0..15]$ into $W[16..63]$ using

$$W[t] = \sigma_1(W[t-2]) + W[t-7] + \sigma_0(W[t-15]) + W[t-16] \pmod{2^{32}}.$$

The SHA-256 schedule functions are

$$\sigma_0(x) = \mathrm{ROTR}^7(x) \oplus \mathrm{ROTR}^{18}(x) \oplus \mathrm{SHR}^3(x),$$

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$$\sigma_1(x) = \mathrm{ROTR}^{17}(x) \oplus \mathrm{ROTR}^{19}(x) \oplus \mathrm{SHR}^{10}(x).$$

The lag set is

$$L = \{2,7,15,16\}.$$

The seed width is

$$N = 16.$$

Each expanded word is a contact point between a forward index t and a backward lag field:

$$t \mapsto \{t-2, t-7, t-15, t-16\}.$$

Therefore the schedule is not only forward diffusion. It is a forward execution that continually reads backward.

6. Aperture Complement Law

For a recurrence with seed width N and lag set L, define the aperture complement

$$A_N(L) = \{N - \ell : \ell \in L\}.$$

For SHA-256:

$$A_{16}(\{2,7,15,16\}) = \{14,9,1,0\}.$$

Sorted:

$$\boxed{A = \{0,1,9,14\}.}$$

These are the seed coordinates that first become visible at the expansion boundary.

The first expanded word is

$$W[16] = \sigma_1(W[14]) + W[9] + \sigma_0(W[1]) + W[0].$$

Thus the first aperture parent set is

$$P_{16} = \{0,1,9,14\}.$$

The four coordinates are not arbitrary. They are the direct readout of the lag complement.

7. Direct-Entry Spectrum

For seed word $W[j]$, $0 \le j \le 15$, direct appearances in the expansion occur when one of the recurrence slots equals j:

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$$t-2=j,\quad t-7=j,\quad t-15=j,\quad t-16=j,$$

with $t \geq 16$.

Therefore the direct-entry set is

$$D(j)=\{j+16\}\cup\{j+15: j\geq 1\}\cup\{j+7: j\geq 9\}\cup\{j+2: j\geq 14\}.$$

The entry cardinality is

$$|D(j)|=\begin{cases}1, & j=0,\\ 2, & 1\leq j\leq 8,\\ 3, & 9\leq j\leq 13,\\ 4, & 14\leq j\leq 15.\end{cases}$$

Thus the seed window partitions as

$$0 \mid 1 \ldots 8 \mid 9 \ldots 13 \mid 14 \ldots 15.$$

The boundary $j=9$ is structural because

$$9+7=16.$$

It is the first seed word that can enter the expansion zone through the raw $t-7$ slot.

8. The Root-Cover Axis Theorem

The aperture complement identifies P_{16}. The next expanded word gives the odd-chain root:

$$W[17]=\sigma_1(W[15])+W[10]+\sigma_0(W[2])+W[1].$$

Hence

$$P_{17}=\{1,2,10,15\}.$$

Theorem 8.1 - Root-Cover Axis

The unique seed coordinate shared by the first even-chain root $W[16]$ and the first odd-chain root $W[17]$ is $W[1]$.

Proof. Compute the intersection:

$$P_{16}\cap P_{17}=\{0,1,9,14\}\cap\{1,2,10,15\}=\{1\}.$$

Therefore $W[1]$ is the unique shared seed parent of both expansion roots. ▫

8.1 Direct Root Coverage

For $j=1$,

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$$D(1) = \{1 + 15, 1 + 16\} = \{16, 17\}.$$

Thus $W[1]$ directly enters both $W[16]$ and $W[17]$.

These are consecutive expansion words. Therefore, in the contamination recurrence,

$$C_{16} = 1, \qquad C_{17} = 1.$$

Since a consecutive dirty pair exists at the first expansion boundary, saturation is immediate:

$$S_1 = 16.$$

The clean echo set is empty:

$$\boxed{\mathcal{C}_1 = \emptyset.}$$

This proves that $W[1]$ is not merely a hinge vertex. It is the universal root-cover axis and global pinch point of the schedule.

9. Even/Odd Root Chains

The first two expansion roots divide the recurrence into initial parity chains:

$$W[16] \to W[18] \to W[20] \to W[22] \to \cdots,$$

$$W[17] \to W[19] \to W[21] \to W[23] \to \cdots.$$

The parent sets are

$$P_{16} = \{0, 1, 9, 14\},$$

$$P_{17} = \{1, 2, 10, 15\}.$$

Thus the structural hierarchy is

```
W[1] = universal root-cover axis
  |
  |-- W[16] even root
  |      parents: {W[0], W[1], W[9], W[14]}
  |      even chain: 16,18,20,22,...
  |
  `-- W[17] odd root
         parents: {W[1], W[2], W[10], W[15]}
         odd chain: 17,19,21,23,...
```

This resolves the role separation:

$$\boxed{W[1] = \text{global root-cover axis.}}$$

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$$\boxed{W[9] = \text{local face-selection hinge.}}$$

10. W[9] as First Triple-Entry Face-Selection Hinge

For $j = 9$,

$$D(9) = \{16,24,25\}.$$

The three direct entries are:

$$W[16] = \sigma_1(W[14]) + W[9] + \sigma_0(W[1]) + W[0],$$

$$W[24] = \sigma_1(W[22]) + W[17] + \sigma_0(W[9]) + W[8],$$

$$W[25] = \sigma_1(W[23]) + W[18] + \sigma_0(W[10]) + W[9].$$

Thus $W[9]$ enters as

$$\boxed{\{\text{raw}, \sigma_0, \text{raw}\}}$$

at

$$\boxed{t = \{16,24,25\}}.$$

$W[25]$ is the triple convergence point because three of its four inputs carry $W[9]$ contamination:

- $W[23]$ carries the slow arm from $W[16]$.
- $W[18]$ carries the fast arm from $W[16]$.
- $W[9]$ re-enters directly through the $t - 16$ slot.

The fourth input, $W[10]$, is clean relative to $W[9]$ at that point.

11. Contamination Recurrence and Clean Echo Sets

For a fixed seed j, define

$$C_t^{(j)} = 1$$

if $W[t]$ depends on $W[j]$, and

$$C_t^{(j)} = 0$$

otherwise.

The seed condition is

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$$C_j^{(j)} = 1,$$

$$C_k^{(j)} = 0 \quad (k \neq j,\ 0 \leq k \leq 15).$$

The contamination recurrence is

$$C_t^{(j)} = C_{t-2}^{(j)} \vee C_{t-7}^{(j)} \vee C_{t-15}^{(j)} \vee C_{t-16}^{(j)}.$$

The clean echo set is

$$\mathcal{C}_j = \{t \in \{16, \ldots, 63\} : C_t^{(j)} = 0\}.$$

The saturation threshold S_j is the first index s such that

$$C_s^{(j)} = 1, \qquad C_{s+1}^{(j)} = 1.$$

12. Saturation Theorem

Theorem 12.1. If there exists s such that

$$C_s = 1, \qquad C_{s+1} = 1,$$

then

$$C_t = 1 \quad \text{for all } t \geq s.$$

within the forward expansion after the pair appears.

Proof. Since $C_s = 1$, the recurrence forces

$$C_{s+2} = 1$$

through the $t - 2$ slot. Since $C_{s+1} = 1$, the recurrence forces

$$C_{s+3} = 1.$$

By induction, if $C_u = 1$ and $C_{u+1} = 1$, then $C_{u+2} = 1$ and $C_{u+3} = 1$. Therefore all subsequent terms are contaminated. ▫

13. Seed Echo Taxonomy

The complete seed echo taxonomy is:

Seed j	Entry Class	$D(j)$	$\mathcal{C}_j$	$\lvert\mathcal{C}_j\rvert$	S_j	First Dirty Pair
0	single	$\{16\}$	$\{17,19,21\}$	3	22	$(22,23)$

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Seed j	Entry Class	$D(j)$	$\mathcal{C}_j$	$\lvert\mathcal{C}_j\rvert$	S_j	First Dirty Pair
1	double	$\{16,17\}$	$\emptyset$	0	16	$(16,17)$
2	double	$\{17,18\}$	$\{16\}$	1	17	$(17,18)$
3	double	$\{18,19\}$	$\{16,17\}$	2	18	$(18,19)$
4	double	$\{19,20\}$	$\{16,17,18\}$	3	19	$(19,20)$
5	double	$\{20,21\}$	$\{16,17,18,19$	4	20	$(20,21)$
6	double	$\{21,22\}$	$\{16,17,18,19$	5	21	$(21,22)$
7	double	$\{22,23\}$	$\{16,17,18,19$	6	22	$(22,23)$
8	double	$\{23,24\}$	$\{16,17,18,19$	7	23	$(23,24)$
9	triple	$\{16,24,25\}$	$\{17,19,21\}$	3	22	$(22,23)$
10	triple	$\{17,25,26\}$	$\{16,18,20,22$	4	23	$(23,24)$
11	triple	$\{18,26,27\}$	$\{16,17,19,21$	5	24	$(24,25)$
12	triple	$\{19,27,28\}$	$\{16,17,18,20$	6	25	$(25,26)$
13	triple	$\{20,28,29\}$	$\{16,17,18,19$	7	26	$(26,27)$
14	quad	$\{16,21,29,30$	$\{17,19\}$	2	20	$(20,21)$
15	quad	$\{17,22,30,31$	$\{16,18,20\}$	3	21	$(21,22)$

The important edge cases are:

$$\mathcal{C}_1 = \emptyset,$$

so $W[1]$ is the unique zero-echo seed, and

$$\mathcal{C}_0 = \mathcal{C}_9 = \{17,19,21\},$$

although $W[0]$ and $W[9]$ reach that echo set through different structural routes.

14. Route-Equivalence at $S = 22$

A saturation equivalence class appears at

$$S_0 = S_7 = S_9 = 22.$$

The mechanisms differ:

1. $j = 0$ is single-entry. It enters at $W[16]$, propagates through the fast arm to 22, and through the slow $t - 7$ arm to 23.
2. $j = 7$ is double-entry. Its direct entries are already $\{22,23\}$.
3. $j = 9$ is triple-entry. Its first entry is 16, and the fast and slow arms collide at $(22,23)$.

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Therefore S_j does not merely count entry cardinality. It identifies route convergence into a shared saturation address.

15. Fold-State Vector and Face Selection

$W[9]$ occupies bytes 36,37,38,39 of a SHA-256 final block. Let r be the number of content bytes in that final block, with $0 \le r \le 55$.

Define

$$c_9(r) = \min\big(4, \max(0, r - 36)\big),$$

$$p_9(r) = \begin{cases} 1, & 36 \le r \le 39, \\ 0, & \text{otherwise}, \end{cases}$$

$$z_9(r) = 4 - c_9(r) - p_9(r).$$

The fold-state vector is

$$\Theta_9(r) = \frac{1}{4}\big(c_9(r), p_9(r), z_9(r)\big).$$

The three occupancy faces are:

$$r < 36: \quad \Theta_9(r) = (0,0,1),$$

$$36 \le r \le 39: \quad \Theta_9(r) = \text{padding crease},$$

$$r \ge 40: \quad \Theta_9(r) = (1,0,0).$$

Thus $W[9]$ is a length-sensitive face selector.

16. The $W[16]$ Divergence Law

When $W[14] = 0$, the first aperture simplifies to

$$W[16] = W[0] + \sigma_0(W[1]) + W[9] \pmod{2^{32}}.$$

If $W[0]$ and $W[1]$ are fixed between two trials a and b, then

$$\boxed{\big(W[16]^{(a)} - W[16]^{(b)}\big) \bmod 2^{32} = \big(W[9]^{(a)} - W[9]^{(b)}\big) \bmod 2^{32}.}$$

This is an exact pre-avalanche diagnostic. The final digest is a still. The $W[16]$ differential is the stream trace.

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17. Sixteen-Gon Chord Geometry

The 16-word seed window can also be represented as a 16-gon. For vertices $a, b \in \{0, \ldots, 15\}$, define the circular distance

$$\delta(a, b) = \min(|a - b|, 16 - |a - b|).$$

Define the central chord angle

$$\phi(a, b) = \frac{360^\circ}{16}\delta(a, b) = 22.5^\circ\delta(a, b).$$

Then the key aperture angles are:

Vertices	δ	Exact angle	Role
$0 \leftrightarrow 14$	2	45°	resting/seam offset
$1 \leftrightarrow 14$	3	67.5°	observer chaos tilt
$0 \leftrightarrow 9$	7	157.5°	genesis-to-hinge alignment
$1 \leftrightarrow 9$	8	180°	diameter / root hinge line
$0 \leftrightarrow 1$	1	22.5°	lean scale
$9 \leftrightarrow 14$	5	112.5°	supplement arm

This gives a derivation path for the previously observed approximate angles:

$$44.35^\circ \approx 45^\circ,$$

$$65^\circ \approx 67.5^\circ,$$

$$156^\circ \approx 157.5^\circ.$$

The exact values belong to the 16-gon chord model. Any empirical projection angle should be compared against these discrete central-angle values rather than asserted without a projection functional.

18. Transformer Attention as Counter-Rotating Aperture Contact

A causal transformer computes attention as

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top + M}{\sqrt{d_k}}\right)V.$$

Here:

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- Q is the local query aperture.
- K, V are stored shape fields.
- M is the architectural mask.
- The token stream moves forward.
- The attention field reaches backward through prior context.

Thus a token is not interpreted as an isolated still. It is resolved by counterflow contact against context.

18.1 Forbidden Seam vs Balance Seam

A causal mask imposes a hard structural seam:

$$M_{ij} = -\infty \quad \text{for } j > i.$$

This makes future positions unavailable to token i. It is a **forbidden seam**.

A balance surface such as

$$q_i \cdot k_j = 0$$

or

$$q_i \cdot k_j = q_i \cdot k_\ell$$

is different. It is not hard zero attention by itself, because

$$\mathrm{softmax}(0)$$

still assigns finite weight unless masked or dominated by competing logits. Such surfaces are **balance seams,** not forbidden seams.

This distinction is essential. The mask is structural exclusion. Dot-product balance is dynamic equilibrium.

19. Language as Stream Crossing

A word token is a still:

$$w_t.$$

Meaning is not contained in the isolated token. Meaning appears where the token crosses the context aperture:

$$\mu_t = \alpha(w_t, C_{\le t}, G, M).$$

For example, the same still word can pass through different context fields:

$$\text{bank}_{\text{river}}, \qquad \text{bank}_{\text{finance}}, \qquad \text{bank}_{\text{aircraft}}.$$

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The token is the same local object. The stream crossing differs. Therefore:

> The word is the still. Meaning is the stream crossing it.

This is the language-processing version of the SHA principle: the output still hides the recurrence, but controlled perturbation reveals the aperture.

20. SHA-to-Transformer Transport Hypothesis

The following table gives the proposed transport map. It is a research hypothesis, not a completed equivalence proof.

SHA-256 Structure	Transformer Structure	Functional Role
Seed window $W[0..15]$	Prompt/context window	Initial symbolic surface
Lag set $\{2,7,15,16\}$	Attention reach pattern and layer connectivity	Backward reference field
Aperture complement $\{0,1,9,14\}$	Structural activation thresholds	Entry gates
$W[1]$	Global context/root carrier	Root-cover axis
$W[9]$	Query-like local hinge	Face selection / dynamic perturbation
$W[14]$	Collapsed seam / boundary state	Constraint surface
$W[15]$	Position/mask/length boundary	Observer boundary
Clean echo set $\mathcal{C}_j$	Attention nulls / low-sensitivity nodes	Negative geometry
Saturation threshold S_j	Layer/position at which perturbation becomes global	Collapse point
Digest/output token	Projected still	Shadow/readout

21. Testable Program

The CRAR model suggests three test families.

21.1 SHA Perturbation Tests

For each seed word j, compute

$$D(j), \quad \mathcal{C}_j, \quad S_j.$$

Verify the taxonomy by direct recurrence. This is already exact in the Boolean contamination model.

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21.2 SHA Projection Tests

For $r = 0, \ldots, 55$, compute

$$\Theta_9(r), \quad W[16](r), \quad W[24](r), \quad W[25](r), \quad h(r).$$

Compare measured projection signatures against the 16-gon chord values

$$45^\circ, \quad 67.5^\circ, \quad 157.5^\circ, \quad 180^\circ.$$

21.3 Transformer Perturbation Tests

For a causal transformer, perturb token j and measure layer-position sensitivity:

$$R_{t,j}^{(\ell)} = \left\| \frac{\partial h_t^{(\ell)}}{\partial x_j} \right\|.$$

Define a thresholded contamination state

$$C_{t,j}^{(\ell)} = \mathbf{1}\left[R_{t,j}^{(\ell)} > \tau \right].$$

Then test whether clean echo-like pockets and saturation-like thresholds exist across layers.

A hard causal clean zone is guaranteed for future-to-past influence:

$$\frac{\partial h_t^{(\ell)}}{\partial x_j} = 0 \quad \text{when } j > t$$

in a strictly causal architecture. The open question is whether internal attention heads create nontrivial negative-geometry pockets analogous to SHA clean echo sets.

22. Discussion

The central inversion is that processors need not move through information. Information can move through a fixed aperture. A model, hash schedule, compiler, or observer can remain structurally fixed while the input surface slides beneath it.

This turns language processing into relative phase contact. The prompt stream moves forward; the model field reaches backward through attention; the output token is the projected residue. The same form appears in SHA-256: the schedule moves forward, but every new word reads backward through fixed lags.

The root-cover result sharpens the SHA side. $W[9]$ is not the whole axis. $W[9]$ is the first triple-entry hinge and length-sensitive face selector. $W[1]$ is deeper: the only seed word that directly enters both first expansion roots and leaves no clean echo.

This produces a refined hierarchy:

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Aperture complement → Root-cover axis → Face hinge → Echo/saturation → Projection.

23. Conclusion

CRAR provides a formal language for a simple operational fact:

You cannot find a stream with a still.

The stream is the recurrence, not the final output. The processor is the aperture, not the moving object. The output is a projection shadow, not the full trace.

In SHA-256, the recurrence exposes the aperture complement $\{0,1,9,14\}$, the root-cover axis $W[1]$, the face-selection hinge $W[9]$, the clean echo taxonomy, and the saturation landscape. In transformer language models, the same mathematics appears as prompt tokens crossing a fixed weight and mask field through attention. The hard seam is the causal mask; the balance seams are equal-score or orthogonality surfaces; meaning emerges at stable contact.

Thus the final sentence of the model is:

The word is the still. Meaning is the stream crossing it.

Appendix A. Key Formula Ledger

A.1 Stream and Perturbation

$$S_{t+1} = F(S_t),$$

$$S_{t+1} = F\left(S_t, S_{t-k_1}, \dots, S_{t-k_m}\right),$$

$$R_F(t,j) = \mathbf{1}\left[\frac{\partial S_t}{\partial S_j} \neq 0\right].$$

A.2 SHA Aperture

$$W[t] = \sigma_1(W[t-2]) + W[t-7] + \sigma_0(W[t-15]) + W[t-16] \pmod{2^{32}},$$

$$L = \{2,7,15,16\},$$

$$A_N(L) = \{N - \ell : \ell \in L\},$$

$$A_{16}(L) = \{0,1,9,14\}.$$

A.3 Direct Entry

$$D(j) = \{j+16\} \cup \{j+15 : j \geq 1\} \cup \{j+7 : j \geq 9\} \cup \{j+2 : j \geq 14\}.$$

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A.4 Root Cover

$$P_{16} = \{0,1,9,14\},$$

$$P_{17} = \{1,2,10,15\},$$

$$P_{16} \cap P_{17} = \{1\},$$

$$D(1) = \{16,17\},$$

$$\mathcal{C}_1 = \emptyset,$$

$$S_1 = 16.$$

A.5 W[9] Hinge

$$D(9) = \{16,24,25\},$$

$$\mathcal{C}_9 = \{17,19,21\},$$

$$S_9 = 22.$$

A.6 Fold State

$$c_9(r) = \min\big(4, \max(0, r - 36)\big),$$

$$p_9(r) = \begin{cases} 1, & 36 \le r \le 39, \\ 0, & \text{otherwise}, \end{cases}$$

$$z_9(r) = 4 - c_9(r) - p_9(r),$$

$$\Theta_9(r) = \frac{1}{4}\big(c_9(r), p_9(r), z_9(r)\big).$$

A.7 Transformer Attention

$$\mathrm{Attn}(Q,K,V) = \mathrm{softmax}\left(\frac{QK^{\top} + M}{\sqrt{d_k}}\right)V,$$

$$M_{ij} = -\infty \quad (j > i).$$

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